Contested Classrooms: Text Analysis of Local School Board Meetings

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Project webpage: https://contested-classrooms.vercel.app

Code: https://github.com/himalbw/SchoolPolitics

Abstract. We perform text analysis of a newly released corpus of school board meetings from 2018–2023 to characterize trends in education politics surrounding the COVID period. Using 80,063 meeting transcripts from 1,485 districts, we combine bag-of-words measures of partisan language, topic models, Higher Criticism tests of vocabulary change and LLM-scored stance. We find that conflict and contested topics such as critical race theory, book content and parental rights rose sharply during the pandemic, but they rose in Republican- and Democratic-leaning districts alike, and in counties with varying experiences with school closures. Partisan differences in language grew only slightly, peaked in 2021, and mostly reverted by 2023. The largest vocabulary divide is between professional and local talk rather than over culture-war issues. The clearest partisan split appears in the positions argued during public comment in late 2021, and it had closed by 2024.

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1 Introduction

Local education politics drew national attention during the COVID period and recovery. Disputes surrounding school closures, mask mandates, and critical race theory (CRT) transformed school board meetings into heated flashpoints representative of the broader culture war (On Point, 2021; CNN, 2021; Pollock et al., 2022). In this paper, we leverage a database of school board meeting transcripts along with text-analysis methods to quantitatively characterize the politicization of education surrounding COVID.

Our dataset DistrictView (Simko et al., 2026; Holman et al., 2026) is a collection of YouTube-captioned school board meeting transcripts, which after screening spans 80,063 meetings from 1,485 districts between 2018 and 2023. Figure 1 summarizes our workflow. To pre-process the transcripts, we restrict to 2018–2023 and to regular school board meetings, tokenize, and drop stop words, and we append county-level partisanship from presidential vote shares as covariates. We then pursue two complementary representations of the text. In the first, we build a bag-of-words representation, reweighted by TF-IDF, and count mentions of a hand-built dictionary of 17 themes: four routine school-business controls and 13 political and contested themes, including two measures of conflict. These themes also guide the LLM analysis: they locate the contested passages we score, and the attributes that large language models score through GABRIEL (Asirvatham et al., 2026b) mirror them. In the second, we fit LDA topic models and apply higher criticism to test whether the vocabulary of meetings changed between the pre-pandemic and post-pandemic periods.

We present four main findings. First, meeting language became more partisan post-pandemic (2021 but cooled by 2023. Second, conflict-driven discussions – such as mentions of critical race theory, book content and parental rights rose in both groups without differing detectably between them. (CHECK THIS)? Third, the trends of topics discussed underwent the most change during COVID, corresponding to a mild temporary polarization between political parties during this period. Fourth, the sharpest divergence came in the positions argued rather than the words used, as LLM-scored conservative positions rose more in Republican-leaning counties in the second half of 2021, mostly in public comment, before the gap closed by 2024.

2 Data and Pre-Processing

We use DistrictView, a collection of U.S. school-governance meeting transcripts from YouTube (Simko et al., 2026; Holman et al., 2026). Our word-based analyses cover 2018–2023.

For the main bag-of-words analysis, we remove unavailable transcripts, transcripts shorter than 1,000 words under its tokenization rules, exact duplicates, and recordings that fail the school-governance eligibility screen. The screen excludes meetings of other public bodies, joint-body meetings, possible clips or campaign events, and unresolved cases. Retained recordings can include regular meetings, special sessions, workshops, and committees. Text is lowercased and converted to word tokens after removing timestamps, bracketed annotations, and punctuation; numerical tokens are excluded from the bag-of-words features.

The resulting corpus contains 80,063 recordings from 1,485 districts in 49 states, totaling approximately 1.13 billion words and 140,000 hours of video. The median recording contains approximately 12,100 words and lasts 1.5 hours. Table 1 summarizes this corpus. HC

uses the same raw data and eligibility rules but a separate preprocessing procedure and a 200-word minimum, producing 83,369 eligible recordings before the matched-panel restrictions in Section 3.2.1. Appendix A documents both sample definitions.

3 Methods

We extend standard notation as discussed in class. Let $\mathcal{D} = \{d_1, \ldots, d_N\}$ denote the corpus, where each document d_n is the cleaned auto-caption transcript of one school-board meeting video, and $N = 80{,}063$ is the number of documents. Each document is a sequence of tokens

$$d_n = (w_{n1}, w_{n2}, \ldots, w_{nL_n}), \qquad w_{n\ell} \in \mathcal{V},$$

where L_n is the length of document n and $\mathcal{V} = \{v_1, \ldots, v_V\}$ is the vocabulary of $V = |\mathcal{V}|$ distinct terms. Each document belongs to a district $i(n) \in \{1, \ldots, I\}$ ($I = 1{,}485$ districts, indexed by NCES LEAID) and a time period $t(n)$ (calendar year or quarter of the meeting date). Each district has a partisan label $P_i \in \{\mathsf{R}, \mathsf{D}\}$, equal to R if its county's 2020 two-party Republican presidential vote share exceeds one half. To avoid a clash with the corpus $\mathcal{D}$, we write the parties in sans-serif and the sets of districts as $\mathcal{I}^{\mathsf{R}}$ and $\mathcal{I}^{\mathsf{D}}$.

3.1 Bag-of-words representation

We represent each document by its bag-of-words count vector $\mathbf{c}_n \in \mathbb{N}^V$, so that the corpus is an $N \times V$ document–term matrix. Because partisanship is measured at the district level, we sum counts within district i and period t, giving $\mathbf{c}_{it}$ with frequencies $\mathbf{q}_{it} = \mathbf{c}_{it}/m_{it}$, where $m_{it} = \sum_j (c_{it})_j$. Each district thus receives equal weight regardless of how many meetings it posts, and the partisanship measure and lexicon rates below are computed from $\mathbf{q}_{it}$.

Following standard practice, we also reweight counts by TF-IDF, which downweights terms that are common to almost every district and emphasizes distinctive ones. For district-period (i, t) and term j,

$$(x_{it})_j = \big(1 + \log(c_{it})_j\big) \left(\log \frac{1 + M}{1 + \mathrm{df}_j} + 1\right),$$

where M is the number of district-periods and df_j is the number that use term j; each row is then normalized to unit length. We use these vectors to find the terms most characteristic of each period and party, and to compare cosine similarity between same-party and different-party pairs of districts.

Dictionary of themes. To measure specific agendas, we define 17 themes by hand, each a list of single words and multi-word phrases (for example, "critical race theory," "book ban" and "point of order"). Four themes cover routine school business and serve as controls; the other 13 cover political and contested themes. Single words are counted from the document–term matrix and phrases directly from the cleaned text, and each theme's rate is its mentions per 10,000 words in $\mathbf{c}_{it}$. Each theme carries a prior about which party discusses it more, which the analysis tests rather than assumes; partisanship always comes from county vote share. The word lists are in the code repository.

3.2 Topic Modeling

If the vocabulary used in school board meetings changes between the periods before and after COVID, it seems plausible that the topics would as well. As such, topic modeling was

applied to the dataset with LDA (Blei et al., 2003). For K topics, the topic model proposes

$$d_i \sim \text{Multinomial}(L_i, \Omega_i), \quad \Omega_i = \sum_{k=1}^{K} \beta_i(k) A_k$$

where $A_k \in \mathbb{R}^p$ is a probability mass function over the p words for topic k so $A_k(j) \geq 0$ and $\sum_j A_k(j) = 1$ and $\beta_i \in \mathbb{R}^K$ is the topic weight vector of document i with $\beta_i(k) \geq 0$ and $\sum_k \beta_i(k) = 1$. Thus, each document's word distribution Ω_i is a mixture of the K topics. To make the model identifiable, LDA assumes that the weights β_i are drawn from a Dirichlet distribution so $\beta_i \overset{iid}{\sim} \text{Dirichlet}(\alpha)$. Equivalently, each word m of document i is generated by drawing β_i, a topic $z_{im} \sim \text{Multinomial}(1, \beta_i)$, and a word $w_{im}|(z_{im} = k) \sim \text{Multinomial}(1, A_k)$.

We use a symmetric prior $\alpha(k) = 0.1$ and a Dirichlet$(0.01 \cdot \mathbf{1}_p)$ prior for each topic. Because the marginal density $\int p(\beta_i; \alpha) p(d_i|\beta_i, A) d\beta_i$ is intractable, we fit the model using variational inference, in which the posterior $(\beta_i, z_{im}|d_i, \alpha, A_k)$ is approximated by factorized Dirichlet and multinomial distributions. We apply the variant of Hoffman et al. (2010). Each chunk of our dataset is treated as a document, and a meeting's topic weights are the average of its chunks' weights, weighted by each chunk's number of in-vocabulary words. We hold out 10% of meetings in a test set for evaluation while the train set is used to estimate vocabulary and topic-word distributions.

3.2.1 Higher Criticism

We test whether mean within-district word shares changed between 2018–2019 and 2022–2023. Starting from the 83,369-recording HC source sample, we require at least four eligible recordings spanning at least three distinct calendar quarters in each of 2018, 2019, 2022, and 2023. This yields 21,878 recordings from 273 districts; coverage in 2020 and 2021 is not required. We retain all eligible recordings in the four comparison years. For each recording, we calculate word shares within a fixed 12,000-term vocabulary, average recordings within district-year, and average the two years within each period. Districts receive equal weight. Support and variance filters leave 11,804 tested terms.

Let $\mathcal{I}_{\text{HC}}$ denote the HC district sample, with $I_{\text{HC}} = |\mathcal{I}_{\text{HC}}| = 273$. For this analysis, $\mathcal{V}_{\text{HC}}$ denotes the fixed 12,000-term vocabulary, and $\mathcal{J}_{\text{HC}}$ indexes its $m = 11{,}804$ tested terms. For district i and vocabulary term v_j, let Δ_{ij} be the late-minus-early word share, using the recording and year averages described above. We calculate

$$\overline{\Delta}_j = \frac{1}{I_{\text{HC}}} \sum_{i \in \mathcal{I}_{\text{HC}}} \Delta_{ij}, \qquad Z_j = \frac{\sqrt{I_{\text{HC}}}\, \overline{\Delta}_j}{s_j},$$

where s_j is the sample standard deviation of Δ_{ij} across districts. The two-sided normal-approximation p-value is $p_j = 2\Phi(-|Z_j|)$, where Φ is the standard normal cumulative distribution function. The global null is zero mean change for every tested term. Ordering the p-values as $p_{(1)} \leq \cdots \leq p_{(m)}$, we calculate

$$\text{HC}^+ = \max \left\{ 0, \max_{\substack{1 \leq k \leq \lfloor m/2 \rfloor \\ p_{(k)} > 1/m}} \sqrt{m}\, \frac{k/m - p_{(k)}}{\sqrt{p_{(k)}(1 - p_{(k)})}} \right\}. \tag{1}$$

An empty maximum is set to zero. We also report $\max_{j \in \mathcal{J}_{\text{HC}}} |Z_j|$ and $\sum_{j \in \mathcal{J}_{\text{HC}}} Z_j^2$.

We calibrate all three global statistics using a Gaussian district-multiplier bootstrap; implementation, assumptions, and sensitivity checks are given in Appendix A.6.

Phrase-based follow-up. Six exploratory dictionaries measure equity/inclusion, race/racism, book restrictions, parents' rights, school choice/vouchers, and closures/reopening. Rates are matches per 10,000 cleaned alphabetic tokens, including stopwords, with the same recording/year/district weighting. We report paired-district mean changes and pointwise 95% t intervals, and apply Holm adjustment across the six categories separately for each specification. A second specification removes entire approximately 1,000-word chunks containing COVID terms or candidate misrecognitions, recalculates rates on the remaining text, and reapplies the coverage requirements. Dictionaries were refined after context inspection; these analyses are exploratory measures of mentions, not stance or causal effects.

3.2.2 Spectral Clustering

Having obtained a topic weight vector β_i for each of the documents, it becomes of interest to investigate how districts cluster based on the topics they discuss and whether political partisanship varies across clusters. To do this, spectral clustering was performed on a matrix $X \in \mathbb{R}^{I \times K}$ where X_{ij} is the weight that district i places on topic j. Row i of X is equal to γ_i^T, where the weight vector $\gamma_i \in \mathbb{R}^K$ is obtained for district i by averaging the weight vectors for the set of documents $\mathcal{S}_i$ that belong to district i: $\gamma_i = \frac{1}{|\mathcal{S}_i|} \sum_{j \in \mathcal{S}_i} \beta_j$.

X is standardized by removing the mean of each column and dividing by the standard deviation within each year to obtain X^*, and singular value decomposition is performed so $X^* = UDV^T$, where orthogonal matrices U and V contain the left and right singular vectors (respectively) and diagonal D the singular values (in non-increasing order). Let X have rank r. Because standardizing removes one degree of freedom, X^* has rank $r-1$, so with noise, the rows of U roughly divide into r clusters. After determining the rank of X^* from a plot of its singular values, k-means clustering into r clusters is performed on the first $r-1$ left singular vectors of X^* and examined for prominent topics and political partisanship per cluster.

3.3 LLM-Based Methods

One important limitation of word-count-based methods is that they do not reveal which side a speaker takes. For example, "mask mandate" is counted the same whether a parent demands or objects to one, and "equity" is the same in a staff report as in a shouted objection. Topic models share the limitation, since a topic is a distribution over words rather than a position. We therefore use a large language model, as in Asirvatham et al. (2026a), to answer focused positional questions about every passage, so that qualitative judgments of the kind a human coder would make are produced at scale and can be compared across districts and years.

GABRIEL (Asirvatham et al., 2026b) is a thin wrapper around a model API: its `rate` function sends each text with a dictionary of attribute definitions in a single prompt and parses the reply into a 0–100 score per attribute. Nothing is learned from our data; the model applies its background knowledge of U.S. political discourse to the rubric we supply, so the rubric wording is the entire specification of the measure. Asirvatham et al. (2026a) report that such scores match human coders across many labeled tasks and are robust to rewording;

Ludwig et al. (2025) note that a model-generated score used as a regression variable should be validated, since a different model or prompt can change estimates. We treat the scores as descriptive measurements and validate them by repeat runs, agreement between two models of different size, and reading (Section B).

Sample and attributes. Because the model reads passages in context, we work from the raw captions rather than the cleaned corpus. We keep 348 districts with at least two eligible recordings in every year of 2018–19 and 2021–23, sample three recordings per district-year (six in 2020, 2021 and 2024, added to locate the timing of any change and test its persistence), and from each of the 10,167 meetings take one random window of about 1,500 words plus up to two windows centered on contested-topic lexicon terms. Random windows measure typical meeting content; flagged windows measure how contested topics are argued once raised, and the two are never pooled. District, place and personal names are replaced with placeholders so the model cannot infer a district's politics from its name. Each excerpt is scored on six attributes in one call: hostility; share of discussion on culture-war issues; parental-rights framing; contention over COVID policy; and the strength with which speakers argue the progressive side, and the conservative side, of a contested dispute. The position attributes count only positions argued in a disagreement, so routine descriptions of district programs score zero. To prevent tuning the attributes to the result, districts are split in half by a deterministic hash of their NCES identifier; wording was developed on a 200-excerpt pilot from split A, revised once, and frozen, and split B is held out to confirm the results. All 28,374 excerpts are scored by `gpt-6-luna`; a separate pass scores the share of each excerpt that is public comment rather than board or staff discussion. Sample construction, attribute definitions, name removal and validation details are in Section B.

Estimation. We average scores to the district-year and, for each attribute, estimate

$$y_{it} = \alpha_i + \gamma_t + \sum_{s \neq 2019} \beta_s \, \mathbf{1}\{t = s\} \, G_i + \mathbf{x}_i' \boldsymbol{\delta}_t + \varepsilon_{it}, \tag{2}$$

where y_{it} is the mean score in district i and year t, G_i is the county 2020 two-party Republican share in tens of percentage points, α_i and γ_t are district and year effects, $\mathbf{x}_i$ contains locale indicators interacted with year, and standard errors are clustered by district. Each β_s measures how the association between county partisanship and the score in year s differs from 2019; district effects restrict identification to within-district change. We re-estimate Eq. (2) at half-year resolution for 2020–21, separately for excerpts above and below 50 on the public-comment share, and with a single post-period indicator (2021–24) for each sample half. We use `statsmodels` 0.15 with package defaults; the design is descriptive, not causal.

4 Results

4.1 Bag-of-Words

We count mentions of the 17 dictionary themes (Section 3.1). Four cover mainstream school business (budget, facilities, athletics and academics) and serve as controls. The other 13 cover political and contested themes: parental rights, book content, critical race theory and race, gender and LGBTQ issues, school choice, COVID mandates, equity and social-emotional

learning, guns and safety, unions and pay, immigration, religion, and two measures of conflict, in-meeting disruption and legal disputes.

Figure 2 documents partisan differences in how often these themes are mentioned in 2022–23. The largest gaps are in professional and local vocabulary rather than culture-war topics: Democratic-county districts use far more equity and social-emotional-learning language ($t = -17.5$), along with more union, pay and academic terms, while Republican-county districts talk more about athletics ($t = 9.8$) and facilities ($t = 6.8$). By contrast, critical race theory, book content, parental rights and school choice do not differ detectably ($|t| < 1.3$).

Figure 3 shows the time trend for six key themes of education politics. COVID mandates and critical race theory spiked in both groups in mid-2021 and faded by 2022. Meeting conflict followed the same path: after dipping in 2020, when many meetings were virtual, it peaked in the third quarter of 2021 and has since fallen back to about its pre-pandemic level. Book content, parental rights and gender issues rose later and stayed above their 2018–19 levels through 2023, in Republican- and Democratic-county districts alike. Figure 4 shows the geography: conflict language rose in many states between 2018–19 and 2022–23, but most in New Jersey, Oklahoma and Florida, which reach about 1.8–1.9 mentions per 10,000 words.

4.2 Topic Modeling

The outputs from LDA were obtained for $K = 10, 20, 30$ topics, with $K = 20$ topics being selected because it achieved the highest coherence score while remaining interpretable (Table 2). Some of the most prominent topics across the entire corpus include informal discussion, voting, budget and finance, and scheduling. Table 3 displays key words and labels for each topic. The trend of these topics over time is examined by averaging the estimated topic weights across meetings within the same district and then across districts within the same year. This analysis is restricted to the 363 districts that were present across all 6 years. Figure 5 displays the trends for each topic, highlighting in color the topics that experienced the most change. Notably, the COVID/Health and parental concerns topics peaked during COVID years (2020-2021) while student activities dropped. In both cases, the topics roughly returned to their pre-COVID levels of prevalence in 2022 and 2023. The topic regarding voting interestingly increased over time, including post-COVID.

4.2.1 Higher Criticism

Recorded word shares differ between 2018–2019 and 2022–2023. HC^+ equals 859.85, compared with a 95th-percentile bootstrap threshold of 11.34. All three global tests return Monte Carlo p-values of 0.0005, the reporting limit. After excluding specified COVID-related, number-format, and suspected artifact words, 11,745 words remain and HC^+ equals 819.30; all three tests again reach the reporting limit.

The phrase analysis finds increases in equity/inclusion, race/racism, book-restriction, and parents' rights mentions (Table 5). All four survive Holm adjustment and persist outside COVID-containing chunks, retaining 273 districts and 21,815 recordings. School-choice/voucher and closure/reopening measures show no detectable change.

4.2.2 Spectral Clustering

The scree plot warranted a selection of 3 clusters, as shown in Figure 6. Figure 7 shows the clustering results, along with the average GOP vote share per cluster. Although the districts do not separate into discrete clusters, the clusters still reveal notable differences in political partisanship, with one leaning Democrat, one Republican, and one neutral. The first left singular vector accounts for most of this distinction, achieving a correlation of 0.466 with GOP vote share driven by topics of planning, equity, and athletics. This separation persists significantly across the years, with a mild polarization during COVID (Figure 8). It is notable, however, that the clusters also tend to align with locale, as the Democratic-leaning cluster is composed of 41.5% urban districts while the Republican-leaning cluster is 25.3% rural. As an example, Figure 9 plots the clustering results for districts in year 2021 on top of a heatmap for county GOP share in 2020. Because geographic proximity to a city tends to correlate with political partisanship, ensuring that these clusters were not simply recovering locale becomes of interest. To this end, the share of variance in GOP vote share explained by cluster membership was compared with its distribution when cluster labels were shuffled within locale (city, suburb, town, rural, for 2,000 permutations). In 2020-2022, clusters explained 15-18% of the variance in partisanship, significantly more than from locale alone (Table 4).

4.3 LLM-Based Methods

Conservative positions were rarely argued before the pandemic, appeared everywhere in 2021, and most in Republican-leaning counties. Figure 10 shows yearly means in random excerpts. In 2018–19 the conservative-positions score averages 0.3 in both Republican- and Democratic-county districts, and 2020 is indistinguishable from the pre-period. In 2021 the score rises to 4.3 in Democratic and 6.8 in Republican counties, declines thereafter, and the two groups meet again at 0.9 in 2024. Parental-rights framing follows the same path. In the event study (Table 6), ten points of county Republican share are associated with 1.14 more points of conservative positions in 2021 (95% CI 0.64–1.63) and 0.54 in 2023 (0.11–0.96), relative to 2019; the 2018, 2020, 2022 and 2024 coefficients all include zero. Collapsing the post-2020 years into a single indicator, the coefficient is +0.45 ($p = .002$) overall and +0.55 ($p = .014$) in the held-out split B; for parental-rights framing it is +0.45 ($p = .001$) and +0.51 ($p = .004$). The gap is not permanent: it opens in 2021, reappears in 2023 as disputes move from masks to library books, and has closed by 2024.

The disagreement and conflict mainly dates to the second half of 2021 and is concentrated in public comment. The half-year coefficients in Table 6 are near zero in both halves of 2020 and small in the first half of 2021. The gap in conservative positions appears in 2021H2 (+1.88, CI 1.05–2.71), the opening of the 2021–22 school year, when mask mandates returned and curriculum disputes peaked; contention over COVID policy spikes in the same half-year (+2.13). The change did not occur while schools were closed. Splitting by who is speaking, excerpts that are mostly public comment show a 2021 coefficient of +12.1 (CI 5.3–18.8) for conservative positions, against +0.42 (0.07–0.78) for excerpts that are mostly board or staff discussion, which look alike across Republican- and Democratic-county districts in every year. Public comment is only 6–10% of random excerpts, so these estimates are imprecise and only 2021 excludes zero.

To make these results more tangibile, we pull quotes from Indian River County, Florida (Republican share 0.61) to illustrate the sequence in verbatim: "please make masks optional and allow parents to make this decision for their children" (September 2020); "you will have opened the floodgates of an indoctrination" (May 2021, on critical race theory); "it's not too late to rescind the vote and follow the laws and statutes that protect children" (March 2022, on library books). Most of the district's 34 verified quotes are public comment (Section B). Indian River County was also a hotbed for Moms for Liberty, a far-right parental rights movement founded in 2021 (Southern Poverty Law Center, 2023).

Hostility rose in both groups Random-excerpt hostility rose from 2019 to 2021 in Democratic (3.7 to 5.2) and Republican (2.5 to 5.2) counties and remained elevated. Republican-county districts were the less hostile group before the pandemic (2019 gap -1.15, CI -2.12 to -0.21) and converged; the post-period coefficient ($+0.39$, $p = .007$) does not replicate in split B ($+0.25$, $p = .16$). Progressive positions show no partisan change. In flagged excerpts (Fig. 11), hostility rose in both groups on every topic except equity, and both groups moved in the same direction on most topics; books and race in the curriculum are the exceptions, with Republican-county meetings well to the conservative side after 2020.

Limitations. There are several important limitations to these results, including the fact we use county vote share as proxies for district partisanship, public-comment share is inferred from caption text, and scores are validated by cross-model agreement and reading rather than human coding. Within those limits the results agree with Section 4.1: culture-war topics rose everywhere at once, and for roughly two years, through public comment rather than board discussion, the positions argued diverged by county politics and then converged.

5 Discussion

We make three main contributions. First, we provide a systematic, multi-method text analysis of DistrictView, a newly released corpus of school board meetings, complementing work that measures conflict in local politics (Holman et al., 2026).

Second, we provide a broad, quantitative analysis of how education politics changed around the pandemic, complementing qualitative evidence. We show that contested topics such as critical race theory, book content and parental rights became salient rhetoric in Republican- and Democratic-leaning districts at the same time, and hostility rose in both. Partisan differences in language grew only slightly and mostly reverted by 2023. School closures account for part of that partisan divergence, but not for the broader rise in conflict, which was shared universally across districts. The clearest partisan split came in late 2021, when conservative positions argued in meetings rose more in Republican-leaning counties, mostly during public comment. Its timing coincides with the rise of parents'-rights organizations such as Moms for Liberty, which future work should consider.

Third, we demonstrate the comparative value of text analysis methods for quantiative social science tasks. Simple bag-of-words measures are surprisingly informative about which topics boards discuss and which words distinguish partisan contexts, but they record mentions rather than positions. Higher Criticism provides a formal hypothesis test allowing us to confirm that meeting vocabulary changed between 2018–19 and 2022–23 across many weak word-level signals. Topic models summarize the meeting agenda and capture some partisan information,

but they are dominated by routine governance and are sensitive to data preparation. LLM scoring captures the positions speakers argue, which word-based methods miss, at the cost of depending on the wording of the scoring rubric and the model used.

Our findings have several limitations. The sample includes only districts that post meetings on YouTube and is not representative of all U.S. districts. County vote share is a proxy for district partisanship. Automatic captions contain transcription errors and almost never identify speakers. Our word-based measures rely mostly on single words, and many topics capture procedural rather than political content. Future work could combine LLM-based measurement with formal hypothesis tests, use district-level election returns, or link changes in meeting content to the formation of local parents'-rights groups.

References

H. Asirvatham, E. Mokski, and A. Shleifer. GPT as a measurement tool. Working Paper 34834, National Bureau of Economic Research, 2026a.

H. Asirvatham, E. Mokski, and A. Shleifer. GPT as a measurement tool. NBER Working Paper 34834, National Bureau of Economic Research, 2026b.

D. M. Blei, A. Y. Ng, and M. I. Jordan. Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022, 2003.

CNN. Critical race theory, COVID rules whip up school board meetings to dismay of students. CNN, republished by WTOP News, Oct. 2021. URL https://wtop.com/national/2021/10/critical-race-theory-covid-rules-whip-up-school-board-meetings-to-dismay-of-students/.

M. D. Hoffman, D. M. Blei, and F. Bach. Online learning for latent dirichlet allocation. 23, 2010.

M. Holman, R. Johnson, and T. Simko. Measuring conflict in local politics. *Urban Affairs Review*, 62(2):303–338, 2026. doi: 10.1177/10780874251355893.

J. Ludwig, S. Mullainathan, and A. Rambachan. Large language models: An applied econometric framework. Working Paper 33344, National Bureau of Economic Research, 2025.

On Point. Why school boards are a nexus of America's culture wars. WBUR and NPR, Oct. 2021. URL https://www.wmra.org/2021-10-15/why-school-boards-are-a-nexus-of-americas-culture-wars.

M. Pollock, J. Rogers, A. Kwako, A. Matschiner, R. Kendall, C. Bingener, E. Reece, B. Kennedy, and J. Howard. The conflict campaign: Exploring local experiences of the campaign to ban "critical race theory" in public K–12 education in the U.S., 2020–2021. Technical report, UCLA Institute for Democracy, Education, and Access, Jan. 2022. URL https://idea.gseis.ucla.edu/publications/the-conflict-campaign/.

T. Simko, R. Johnson, M. Holman, and M. Reece. DistrictView, 2026.

Southern Poverty Law Center. Moms for liberty. Extremist Files, https://www.splcenter.org/resources/extremist-files/moms-liberty/, 2023. Accessed November 5, 2026.

Figures

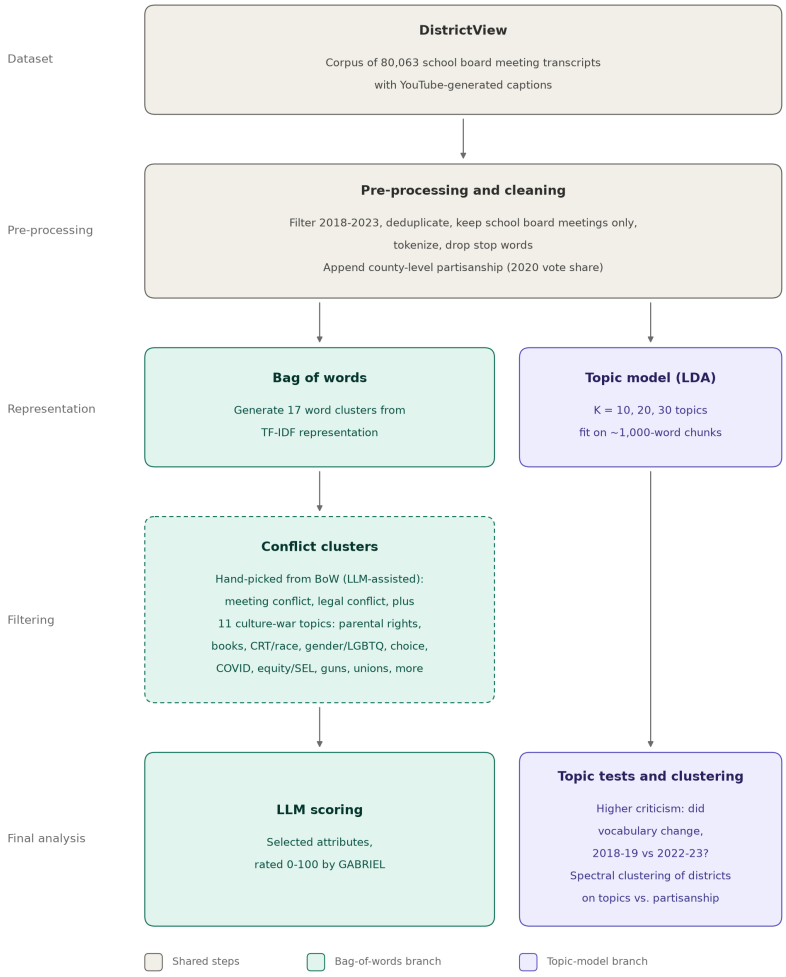


Figure 1: Project workflow. DistrictView transcripts are pre-processed and represented as bag of words (with TF-IDF) and LDA topic models. A hand-built dictionary of 17 themes is counted on the bag of words and guides the LLM scoring; the topic model feeds the higher-criticism test and the spectral clustering of districts.

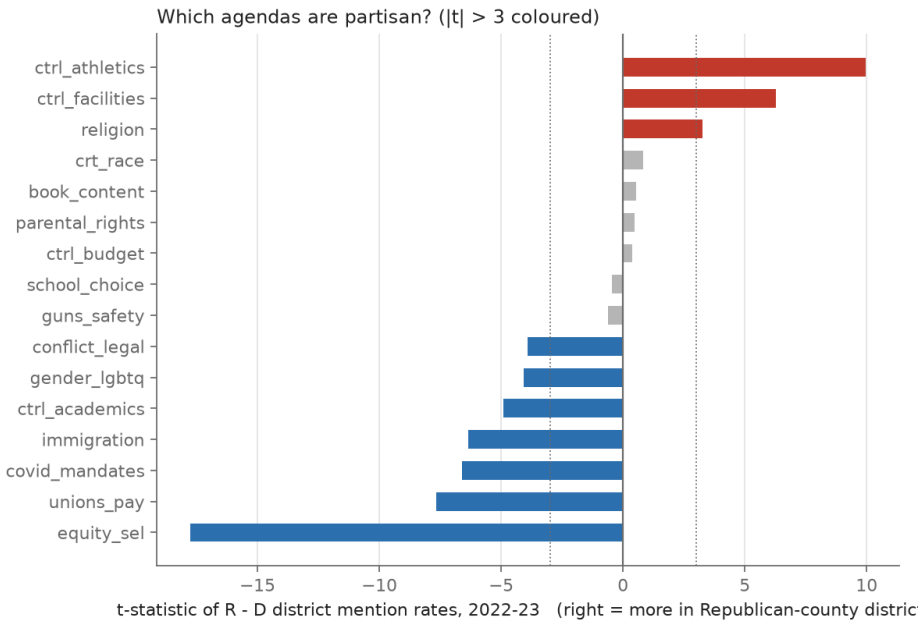


Figure 2: Partisan differences in contested-topic and control vocabulary, 2022–23. Bars show Welch t-statistics for the difference in district mention rates per 10,000 words between Republican- and Democratic-county districts (positive = more common in Republican counties). Categories with $|t| > 3$ are colored; dotted lines mark ± 3. Categories prefixed `ctrl` are non-contested controls. No locale controls.

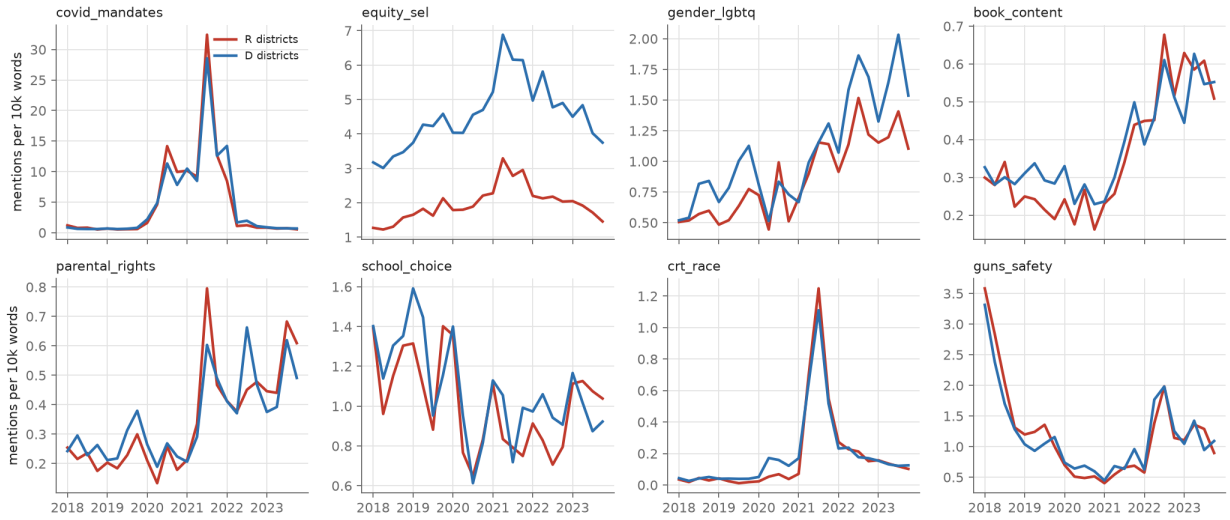


Figure 3: Mentions of six themes of education politics over time. Quarterly mean of district mention rates per 10,000 words in Republican-county (red) and Democratic-county (blue) districts, 2018–2023; the grey band marks 2020–21. Vertical scales differ across panels. Meeting conflict counts decorum and disruption terms (e.g. "out of order", "point of order"). Mentions count words used, not the position taken.

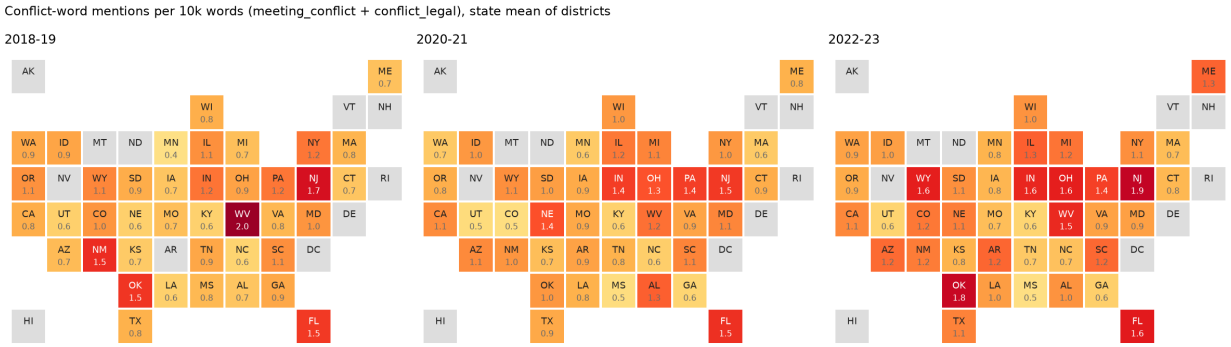


Figure 4: Conflict language by state and period. Each tile shows the state mean of district mention rates per 10,000 words for meeting-conflict terms (e.g. "out of order", "point of order") and legal-conflict terms (e.g. lawsuit, resignation, censure). States with fewer than five sample districts are grey. Captions carry no tone, so this measures conflict that is said, not heard.

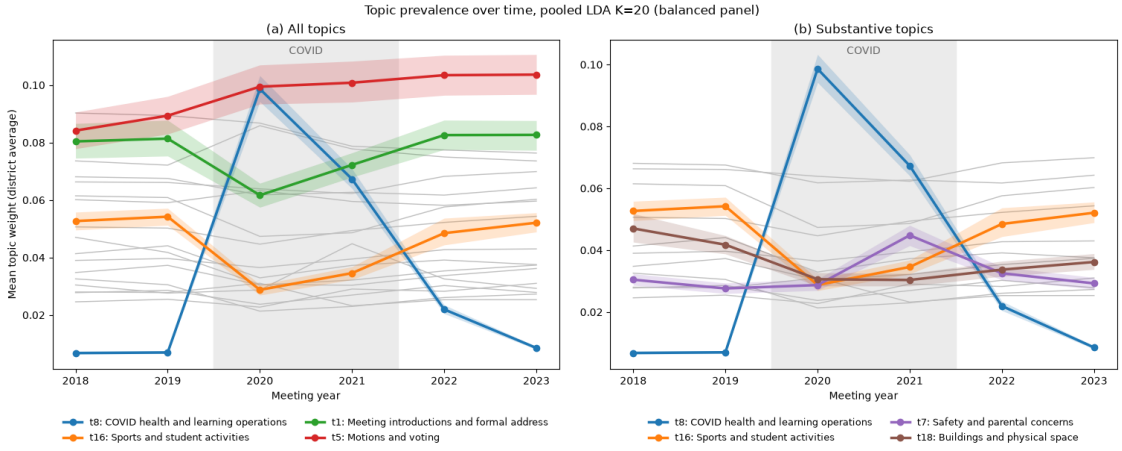


Figure 5: Topic prevalence over time, pooled LDA ($K = 20$), balanced panel of 363 districts observed in every year 2018–2023. Lines show the mean topic weight, averaged over meetings within each district and then across districts. Colored lines highlight the four topics with the largest change from the 2018–19 baseline. Shaded bands are 95% bootstrap intervals from resampling districts. Gray lines are the remaining topics; the gray band marks 2020–21. (a) All 20 topics. (b) Substantive topics only, excluding the five procedural and conversational topics (T1, T3, T4, T5, T20).

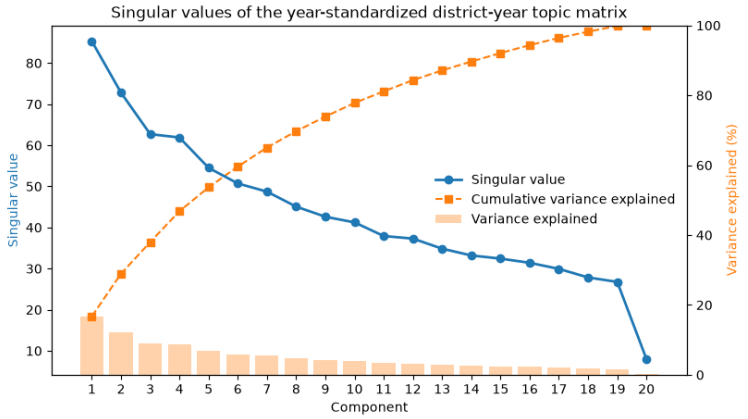


Figure 6: Singular values (left axis) and percent of variance explained (right axis) of the year-standardized district-year topic matrix. We keep rank $r = 2$ and thus cluster districts into $r + 1 = 3$ groups. The third and fourth singular values are nearly equal, which makes their singular vectors poorly identified, so small changes in the data can rotate the two directions within the plane they span. Rank 4 (five clusters) would therefore rest on unstable directions, and it would also break the sample into many small groups that are harder to interpret and compare by partisanship. Three clusters keep the groups large and interpretable for the partisanship analysis.

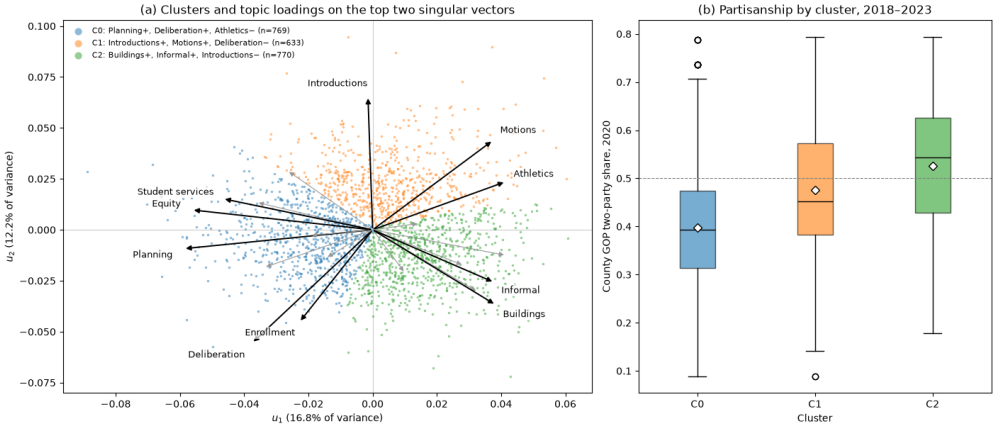


Figure 7: Spectral clustering of district-year topic profiles. (a) Points are district-years plotted on the top two left singular vectors and colored by cluster. Arrows show each topic's correlation with the two component scores, so topics with long arrows define the directions and a cluster lying toward an arrow emphasizes that topic relative to other districts in the same year. Only the ten longest arrows are labeled. (b) County two-party GOP vote share in 2020 by cluster, pooled over all years. Boxes show the interquartile range, black lines mark medians, diamonds mark means, and the dashed line marks 50%.

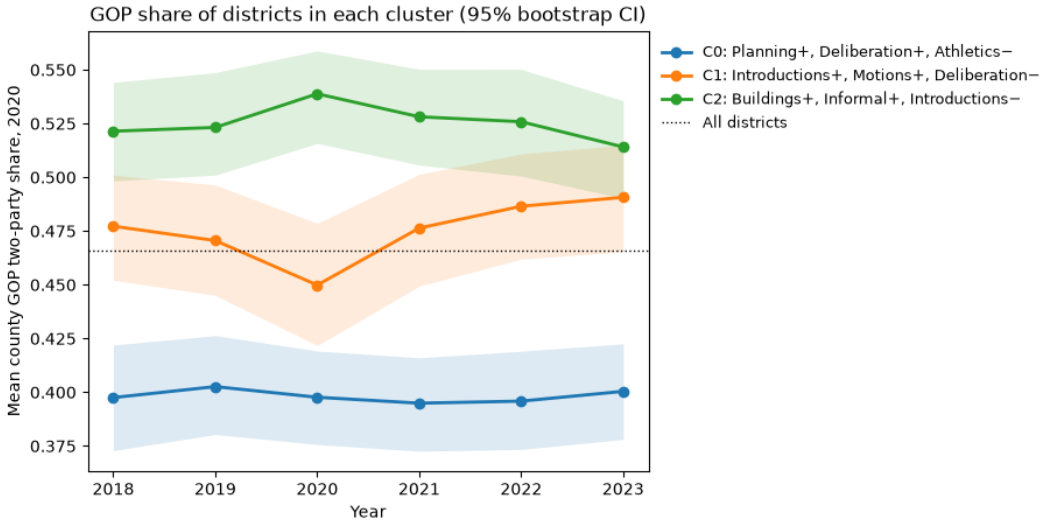


Figure 8: Mean county GOP two-party vote share (2020) of the district-years in each cluster, by year. Each point averages the 2020 GOP share over all districts assigned to that cluster in that year. Shaded bands are 95% bootstrap intervals, in which whole districts are resampled with replacement, keeping all six of a district's years together, the cluster-by-year means are recomputed for each of 500 draws, and the 2.5th and 97.5th percentiles are taken. Cluster assignments are held fixed (not refit). The dotted line is the mean over all districts, each counted once.

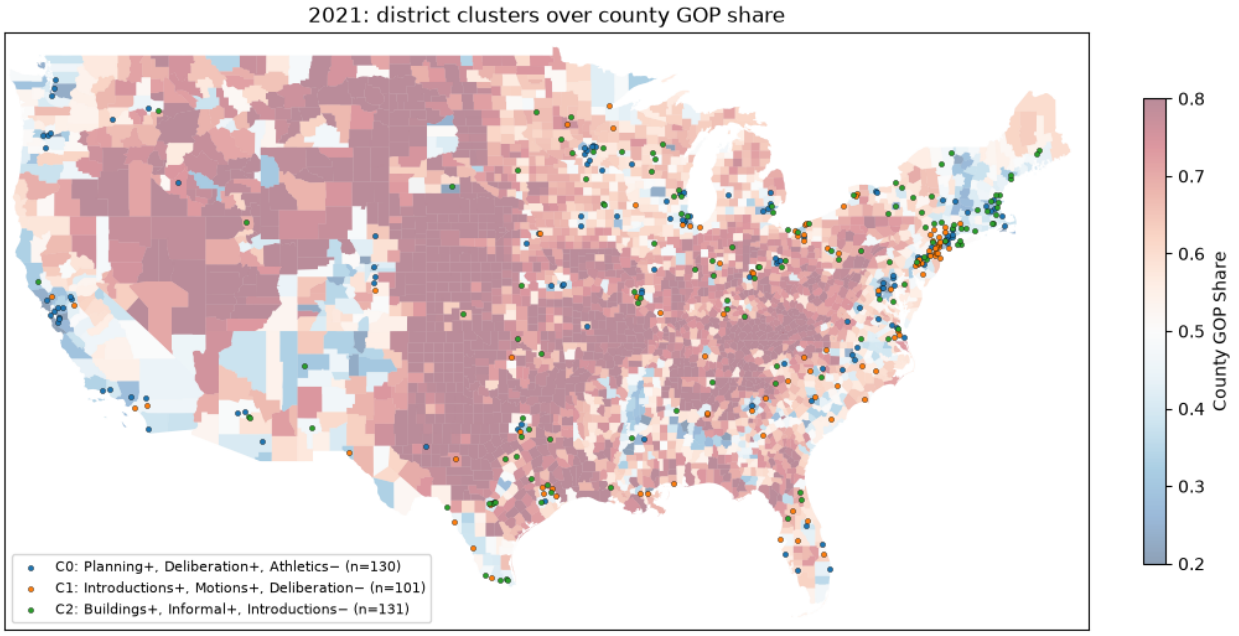


Figure 9: District clusters in 2021. Each point is a district in the balanced panel, placed at its district office and colored by its 2021 cluster assignment. Background shading is the 2020 county two-party GOP vote share (blue = Democratic, red = Republican; grey = no data). Shading is at the county level, so it approximates rather than measures each district's partisanship.

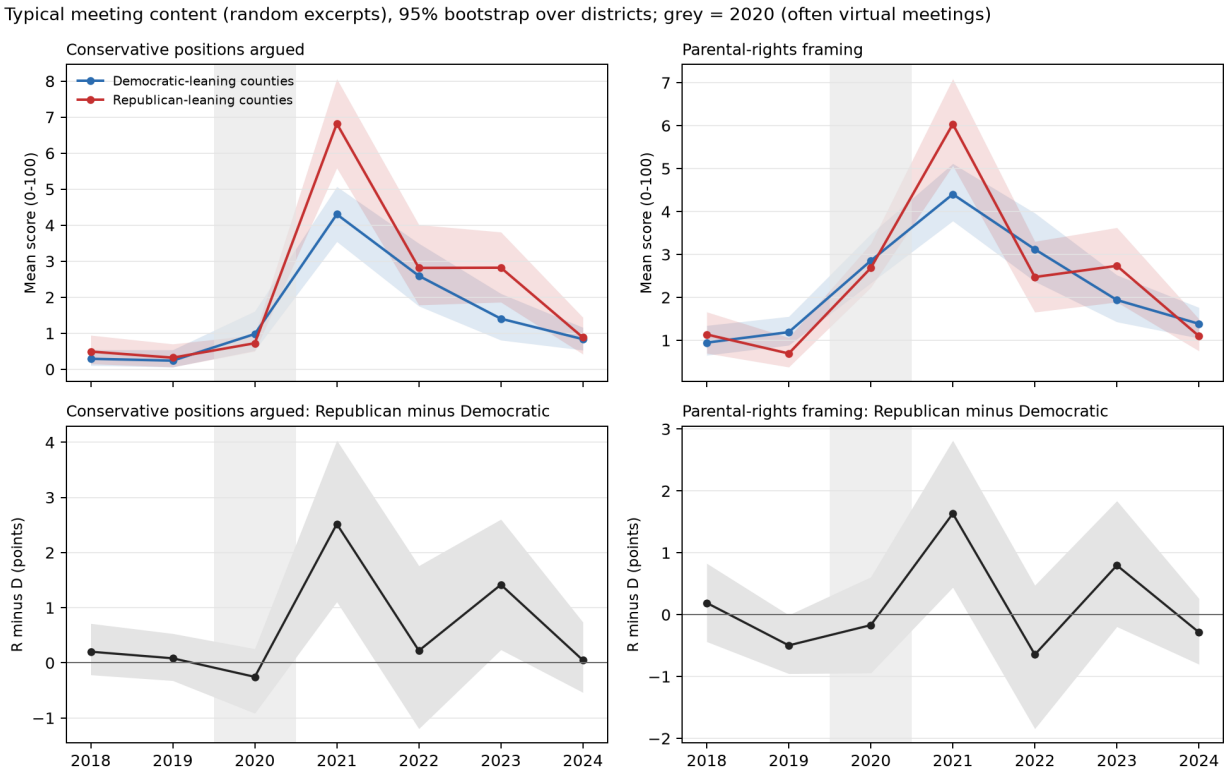


Figure 10: LLM-scored conservative positions and parental-rights framing in random excerpts, 2018–2024. Top: yearly means for districts in Republican- and Democratic-leaning counties, 95% bootstrap bands over districts. Bottom: Republican minus Democratic. Grey marks 2020, when many meetings were virtual. 348 districts, 16,367 random excerpts.

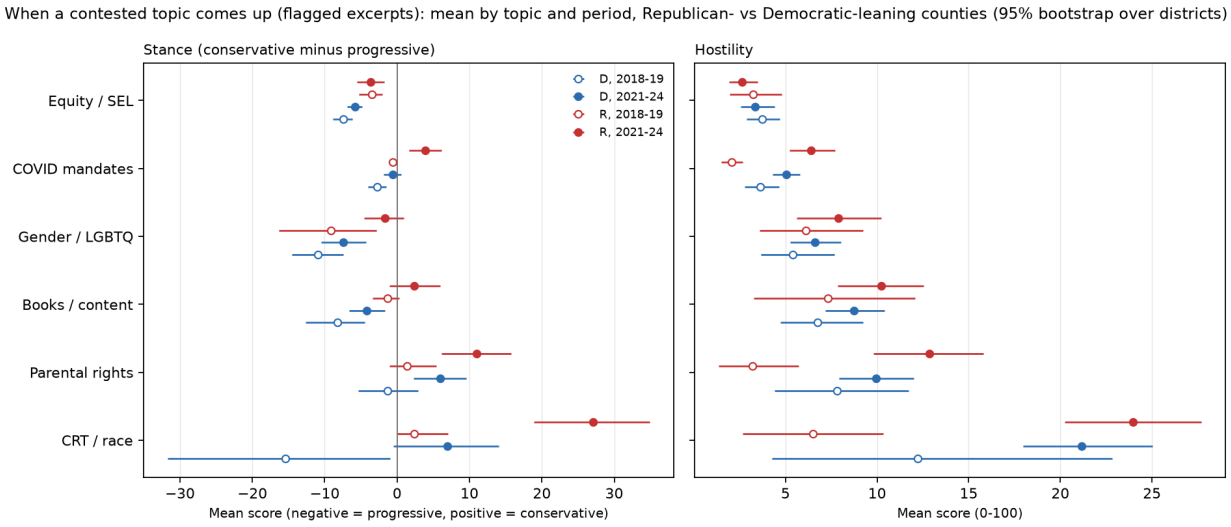


Figure 11: Flagged excerpts: mean stance (conservative minus progressive) and hostility by contested topic, 2018–19 (open) and 2021–24 (filled), Republican- and Democratic-county districts, 95% bootstrap intervals over districts.

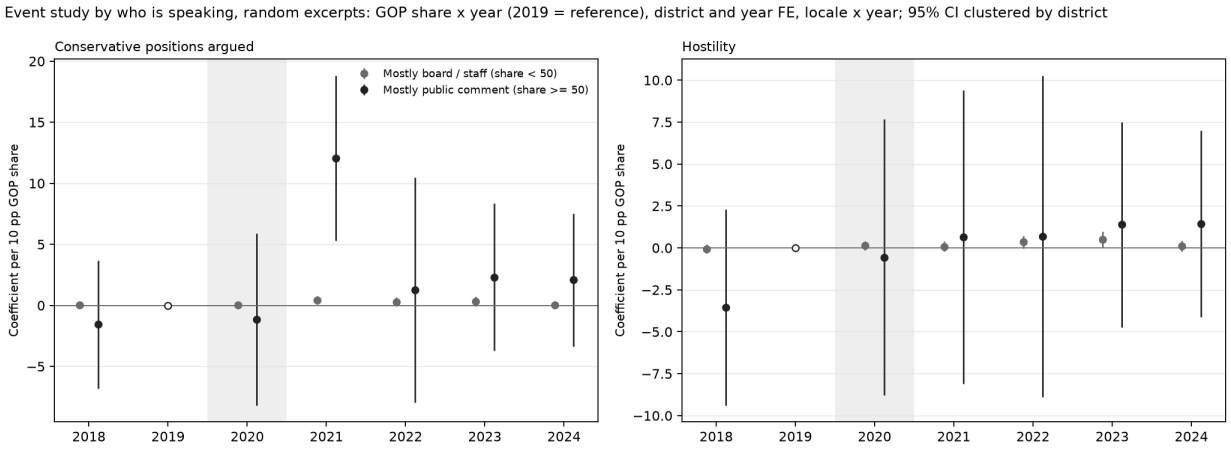


Figure 12: Event study by who is speaking: coefficients on county Republican share × year (2019 reference) for excerpts that are mostly public comment (share ≥ 50) and mostly board or staff discussion, random excerpts, district and year fixed effects, 95% CIs clustered by district. Public-comment cells hold 55–199 excerpts per year.

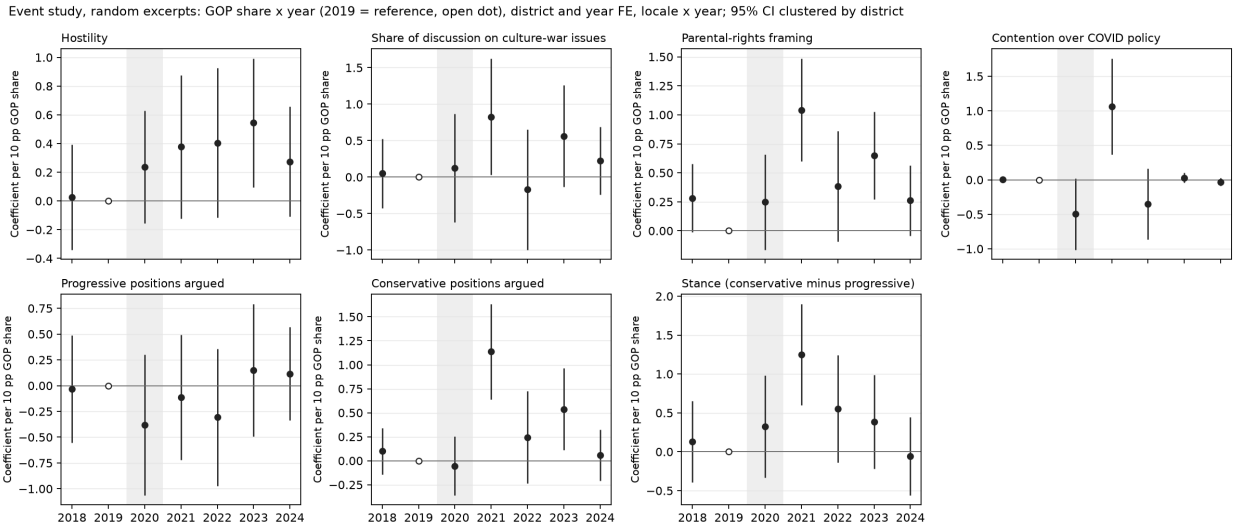


Figure 13: Event study for all seven LLM scores (the six attributes and stance), random excerpts, 2019 reference, district and year fixed effects, locale×year controls, 95% CIs clustered by district.

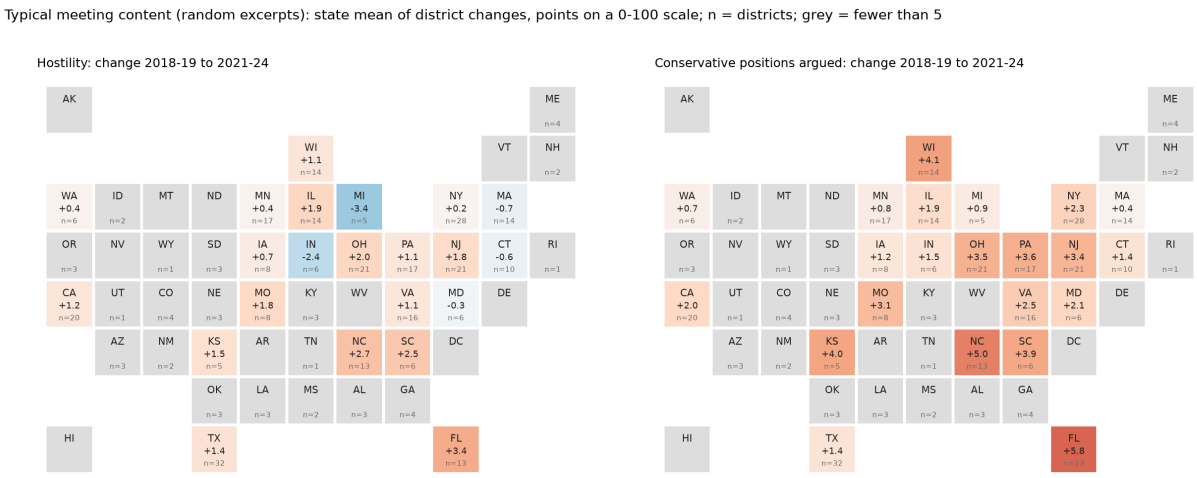


Figure 14: Change in random-excerpt hostility (top) and conservative positions (bottom) from 2018–19 to 2021–24, state means over sampled districts; states with fewer than five districts are grey, and each tile shows the number of districts.

Tables

Table 1: Summary of the analysis corpus.

Group	Districts	Meetings	Words (millions)	Period
Panel A: by county partisanship				
Republican county	703	31,260	380.4	2018–23
Democratic county	779	48,517	743.9	2018–23
Unclassified (Alaska)	3	286	4.9	2018–23
Panel B: by period				
Pre-COVID	545	12,866	161.1	2018–19
COVID	1,277	30,516	453.3	2020–21
Post-COVID	1,339	36,681	514.8	2022–23
Total	1,485	80,063	1,129.1	2018–23

Notes: A meeting is one YouTube video that passes the sample filters. Words are counted in the cleaned transcripts before stopword and vocabulary cuts. Partisanship is the district-office county's 2020 two-party Republican vote share, split at 0.5. Districts in Panel B do not sum to the total because most appear in more than one period.

Table 2: LDA model fit by number of topics K, evaluated on held-out transcript chunks. Bold marks the best value in each column.

Topics (K)	Completion perplexity	Mean NPMI coherence	Top-10 word diversity
10	1,849.08	0.07168	**0.6500**
20	1,754.09	**0.08400**	0.6250
30	**1,709.24**	0.08043	0.6133

Notes: Completion perplexity: each held-out chunk's word counts are randomly split in half, where the topic mixture is inferred from one half and used to predict the other, and perplexity is exp(−log-likelihood/tokens) over all predicted words, so it measures how well the model generalizes to unseen text. *Mean NPMI coherence*: for each topic, the average normalized pointwise mutual information of all pairs of its top ten words, based on how often they appear together in held-out chunks, averaged over topics. It measures whether a topic's top words actually appear together, an indication for human interpretability. *Top-10 word diversity*: the number of distinct words across all topics' top-ten lists divided by $10K$, measuring how little the topics overlap. We use $K = 20$. It has the highest coherence, and while $K = 30$ achieves lower perplexity, the gain is small (2.6% from 20 to 30 vs. 5.1% from 10 to 20) and diversity keeps falling. On reading the topics, $K = 30$ splits coherent themes into very similar subtopics and isolates off-topic material such as advertisements and caption artifacts, whereas $K = 20$ gives more distinct topics and a manageable number of dimensions for the clustering analysis.

Table 3: Topics from LDA ($K = 20$), 2018–2023. Top words are the 8 highest-probability words in each topic. Share is the mean topic weight, averaged over meetings in each district and then across districts. Top is the percentage of meetings in which the topic has the largest weight of the 20.

	Label	Top words	Share (%)	Top (%)
Panel A: Substantive topics				
T14	School updates & recognition	students, work, great, community, staff, teachers, things, see	6.5	6.6
T9	Budget & finance	budget, year, dollars, fund, money, state, funds, tax	6.4	10.1
T13	Curriculum & instruction	students, grade, teachers, year, learning, math, student, level	5.3	8.0
T10	Board policy	board, policy, think, committee, members, make, no, policies	4.8	4.7
T16	Athletics & activities	high, year, team, students, state, sports, great, student	4.7	3.0
T8	COVID operations	students, learning, staff, day, back, health, virtual, time	4.1	6.7
T18	Buildings & space	building, there's, new, see, space, lot, area, kind	4.0	3.6
T6	Enrollment data	see, students, year, schools, look, percent, number, numbers	3.5	2.0
T2	Strategic planning	plan, district, board, report, year, work, data, goals	3.5	2.1
T17	Student services	students, program, services, programs, schools, support, district, education	3.1	1.1
T7	Safety & parents	children, parents, safety, schools, students, district, child, kids	3.0	2.3
T11	Transport & services	bus, transportation, technology, buses, use, food, year, service	2.9	0.8
T19	Staffing & compensation	teachers, staff, teacher, position, employees, year, positions, district	2.7	0.7
T15	Construction & bonds	project, projects, construction, bond, facilities, work, process, bid	2.3	0.9
T12	Equity & race	students, community, equity, people, district, education, schools, black	2.3	1.7
Panel B: Procedural and conversational topics				
T20	Informal discussion	yeah, think, no, kids, people, back, mean, need	10.4	11.4
T5	Motions & voting	motion, mr, board, item, agenda, discussion, approve, aye	9.8	12.7
T4	Deliberation	think, things, kind, there's, yeah, something, need, lot	7.7	10.7
T3	Scheduling	meeting, next, week, day, board, time, last, year	6.7	3.1
T1	Introductions	board, mr, dr, evening, miss, district, members, students	6.4	7.9

Table 4: Partisanship explained by cluster membership, with a within-locale permutation test. η^2 is the share of variance in 2020 county two-party GOP vote share explained by cluster membership. Locale-only η^2 is the same quantity using the four locale categories (city, suburb, town, rural) as groups. p-values compare cluster η^2 with 2,000 permutations of cluster labels within locale.

Year	District-years	η^2 (clusters)	η^2 (locale only)	Permutation p
2018	362	0.126	0.138	< 0.001
2019	362	0.121	0.138	< 0.001
2020	362	0.176	0.138	< 0.001
2021	362	0.153	0.138	< 0.001
2022	362	0.149	0.138	< 0.001
2023	362	0.120	0.138	< 0.001
All years	2,172	0.137	0.138	< 0.001

Table 5: Changes in policy-related mentions per 10,000 cleaned alphabetic tokens, 2018–2019 to 2022–2023. Brackets give pointwise 95% confidence intervals.

Category	All text	Outside COVID chunks
Equity/inclusion	0.778 [0.513, 1.044]	0.816 [0.545, 1.086]
Race/racism	0.220 [0.152, 0.287]	0.234 [0.162, 0.306]
Book restrictions	0.0557 [0.0381, 0.0733]	0.0571 [0.0391, 0.0752]
Parents' rights	0.0965 [0.0649, 0.1280]	0.0987 [0.0665, 0.1308]
School choice/vouchers	−0.006 [−0.051, 0.039]	−0.012 [−0.053, 0.030]
Closures/reopening	0.017 [−0.002, 0.035]	0.006 [−0.013, 0.025]

Table 6: Event-study estimates, Eq. (2): coefficients per 10 percentage points of county 2020 Republican vote share, relative to 2019, with district and year fixed effects and locale×year controls; 95% confidence intervals clustered by district in brackets. Random excerpts. Lower panel: half-year resolution for 2020–21.

	2018	2020	2021	2022	2023	2024
Conservative positions	0.10 [−0.14, 0.34]	−0.05 [−0.36, 0.25]	**1.14** [0.64, 1.63]	0.24 [−0.24, 0.72]	**0.54** [0.11, 0.96]	0.06 [−0.21, 0.33]
Parental-rights framing	0.28 [−0.01, 0.58]	0.25 [−0.16, 0.66]	**1.04** [0.60, 1.49]	0.38 [−0.09, 0.86]	**0.65** [0.27, 1.03]	0.26 [−0.04, 0.56]
Hostility	0.03 [−0.34, 0.39]	0.24 [−0.16, 0.63]	0.38 [−0.12, 0.88]	0.40 [−0.12, 0.92]	**0.54** [0.10, 0.99]	0.27 [−0.11, 0.66]
Progressive positions	−0.04	−0.32	−0.01	−0.31	0.15	0.12

	2020H1	2020H2	2021H1	2021H2
Conservative positions	−0.12 [−0.42, 0.18]	0.16 [−0.10, 0.43]	0.41 [0.01, 0.82]	**1.88** [1.05, 2.71]
COVID contention	0.13 [−0.17, 0.43]	−1.08 [−2.07, −0.10]	0.07 [−0.60, 0.74]	**2.13** [1.06, 3.21]

A Corpus construction and cleaning

The bag-of-words and Higher Criticism (HC) analyses share raw DistrictView inputs and school-governance eligibility rules, but use different preprocessing and minimum-length thresholds. This appendix distinguishes their sample construction and text representations. The HC results are not obtained by applying additional restrictions to the final 80,063-recording bag-of-words corpus.

A.1 Sample construction

HC source sample. We select 110,777 records with nonmissing meeting dates in 2018–2023. We exclude 9,914 transcript placeholders, 1,216 missing transcript files, 1,627 transcripts with fewer than 200 normalized word tokens, and 94 exact duplicates, leaving 97,926 recordings. Duplicate detection compares normalized word sequences and retains the first encountered copy. The eligibility screen below retains 83,369 recordings. Applying the matched-panel requirements in Section 3.2.1 gives 21,878 recordings from 273 districts.

Main bag-of-words sample. The initial bag-of-words pipeline assigns year from meeting date when available and otherwise from upload date, selecting 122,987 records in 2018–2023. Removing 11,333 transcript placeholders, 1,297 missing files, and 7,338 transcripts shorter than 1,000 words under that pipeline's tokenization leaves 103,019 recordings. We intersect these retained records with the HC eligibility-screened identifiers, which also removes records without meeting dates. This leaves 80,063 recordings from 1,485 districts. The intersection removes 11,098 ambiguous recordings, 9,074 without meeting dates, 2,184 meetings of other bodies, 433 possible clips or campaign events, 76 joint-body meetings, 90 exact duplicates, and one recording below the HC pipeline's 200-word threshold. Because the two pipelines count words differently, the 80,063-recording sample is not obtained simply by imposing a 1,000-word threshold on the 83,369-recording HC sample.

A.2 Text normalization

The bag-of-words pipeline removes caption timestamps and bracketed annotations, lowercases text, standardizes curly apostrophes, replaces punctuation with delimiters while retaining internal apostrophes, and normalizes whitespace. Its initial word count is taken before the later bag-of-words feature exclusions.

The HC pipeline lowercases text, standardizes curly apostrophes, removes bracketed annotations other than timestamps, and expands specified contractions and informal forms. It then extracts sequences of letters a–z, allowing an internal apostrophe. Numerals, punctuation, and timestamps do not enter this word sequence. Consequently, the pipelines can assign different word counts to the same recording. Neither pipeline establishes transcription accuracy through audio validation.

A.3 School-board eligibility screen

Titles must either explicitly identify school-board proceedings or contain a governance term supported by channel or transcript evidence. Channel support requires at least ten explicitly titled school-board recordings, constituting at least half the channel's screened recordings, together with at least three education-related word occurrences covering two distinct terms in the first 600 transcript words. Alternatively, the opening text must explicitly identify a

school governing body. We exclude recordings flagged as possible clips or campaign events, other public-body meetings, joint-body meetings, or unresolved cases. The primary eligible sample contains 83,369 recordings: 70,748 with explicit titles and 12,621 with corroborating evidence. Sampled text reviews supplement these rules; eligibility is not manually verified for every recording.

A.4 Bag-of-words representation

We build a transcript-by-term count matrix after removing stop words, keeping terms that appear in at least 20 transcripts. We then drop numerals, terms shorter than three characters and routine procedural words, merge possessives with their base forms, and drop terms used in fewer than 25 districts. Counts are summed to district-quarter and district-period for the partisanship and dictionary measures (Section 4.1) and reweighted by TF-IDF for the distinctive-term and similarity analyses. The multi-word phrases in the dictionary of themes (Section 3.1) are counted directly from the cleaned text, since the matrix holds single words only.

A.5 Higher Criticism representation

A fixed dictionary expands forms such as "can't" to "cannot" and "gonna" to "going to"; suffix rules expand forms such as "n't" to "not" and "'ve" to "have." These transformations precede alphabetic token extraction. Transcripts are divided into consecutive 1,000-word chunks; a final chunk shorter than 200 words is merged into the preceding chunk. No stemming, lemmatization, or automatic spelling correction is applied.

HC reuses the vocabulary fitted for the original LDA analysis, before the later meeting-eligibility screen. Exact-content hashes assign approximately 90% of the 97,926 retained recordings to training and 10% to holdout; all chunks from a recording share its assignment. Vocabulary fitting uses training chunks from 2018–2023 only. We remove scikit-learn's English stopwords except "no," "not," "nor," and "never," and add specified fillers and contraction fragments. Candidate terms contain at least two letters and occur in at least 25 training chunks and no more than 80% of training chunks. We retain the 12,000 eligible terms with the highest total training counts.

For HC, all chunk counts are recombined within each recording. Shares use the recording's total count within the fixed 12,000-term vocabulary. We retain terms occurring in at least 30 of the 273 districts across the four comparison years, with district-contrast standard deviation greater than 10^{-10}, excluding the ordinal fragments `th`, `nd`, `rd`, and `st`. This leaves 11,804 tested terms. Vocabulary selection and dictionary development used the observed corpus, so the analysis is exploratory rather than an independent confirmation of earlier topic discoveries.

A.6 HC calibration and sensitivity checks

For each bootstrap draw, independently generate $g_i \sim N(0,1)$ for each district $i \in \mathcal{I}_{\text{HC}}$ and calculate

$$Z_j^* = \frac{1}{\sqrt{I_{\text{HC}} - 1}} \sum_{i \in \mathcal{I}_{\text{HC}}} g_i \frac{\Delta_{ij} - \overline{\Delta}_j}{s_j}.$$

One multiplier is shared across all terms within a district, retaining the estimated cross-word covariance. Standard errors remain fixed, and all three global statistics are recomputed for each draw, including the same HC truncation. We use $B = 1{,}999$ draws and random seed 236. For observed statistic T_{obs}, the Monte Carlo p-value is

$$\widehat{p}_T = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{T_b^* \geq T_{\text{obs}}\}}{B + 1}.$$

The minimum reportable value is therefore 0.0005. Calibration is approximate and treats district contrast vectors as independent sampling units; it does not fully accommodate shared state or national shocks.

Eligibility sensitivity checks use an explicit-title-only panel of 216 districts and primary eligibility restricted to those same districts. Other checks omit selected annotation terms or progressively exclude core COVID terms, an expanded pandemic-related list, and that expanded list plus number-format and suspected artifact terms. The last specification retains 11,745 tested terms. These checks remove tested coordinates without changing the original vocabulary-share denominators. They do not constitute complete transcription correction or causal adjustment for the pandemic.

A.7 Phrase-based follow-up

We search cleaned chunks using six dictionaries, separately from the fixed HC vocabulary. Apostrophes are removed for matching. Matches within each category are nonoverlapping, but categories may overlap; matches spanning chunk boundaries can be missed. Rates use all cleaned alphabetic tokens, including stopwords. Recording rates are averaged within district-year, years within period, and districts.

Context inspection motivated removing "book challenge," which can describe reading competitions, and replacing bare "voucher" terms with explicit school-choice and education-voucher expressions. Exact dictionaries and COVID-exclusion lists are retained with the analysis code. These refinements are data-informed, not independently validated classifications.

For each category, we calculate the mean paired-district change, a pointwise 95% t interval, and a two-sided paired t-test p-value. Holm adjustment is applied across the six categories separately for the all-text and restricted-text specifications; the displayed intervals are not multiplicity-adjusted.

The restricted-text specification removes entire chunks containing explicit COVID terms or candidate misrecognitions, then recalculates recording rates and reapplies the panel coverage requirements. It retains all 273 districts and 21,815 of 21,878 recordings, with 269,780,215 of the original 299,650,366 alphabetic tokens remaining. This procedure can remove adjacent non-COVID discussion and retain implicit COVID discussion. Neither specification identifies stance or systematically separates public comment from board and staff speech.

B LLM measurement

Sample. Eligible recordings pass the screen of Section A, have at least 3,000 words, and are unique within district and date (the longest is kept). 349 districts have at least two eligible recordings in every year of 2018, 2019, 2021, 2022 and 2023; 348 have vote share and locale.

The 12 strata (Republican-share tercile × city, suburb, town, rural) are uneven because Democratic rural (5) and Republican urban (11) districts are rare; the sample over-represents suburbs and Democratic-leaning counties. Recordings sampled per year, 2018–2024: 1,032, 1,041, 2,036, 2,060, 1,042, 1,040 and 1,916; half-years in 2020 and 2021 receive three each where available, and 16 districts have no eligible 2024 recording.

Excerpts. Random windows have median length 1,495 words. Flagged windows are centered on clusters of at least two lexicon hits, with rarer categories (CRT, parental rights, books, gender) preferred over COVID and equity when several apply; counts by category (post/pre, original sample): COVID 1,496/334; equity/SEL 974/743; gender/LGBTQ 505/271; books 406/147; parental rights 365/81; CRT/race 338/23. A variance decomposition of random-excerpt conservative positions gave 72.7% within district-year, 21.6% between years within district and 5.7% between districts, so two further random windows per meeting were added for 2021 and 2023, narrowing the confidence intervals of the Republican–Democratic gap in those years by 17–31%. Scored excerpts: 16,367 random and 12,007 flagged; 28,374 in total.

Name removal and leakage. District, city, county, state and school names from the metadata, the 50 state names, titles followed by up to two name tokens, phrases ending in "High School," "Elementary," "School District" and similar, and spaCy `en_core_web_sm` `PERSON`/`GPE` entities are replaced by `[NAME]` or `[PLACE]`; "DEI" and "gender" are protected from the tagger, and "I'm X" is deliberately not stripped. Removal deletes 0.1% of lexicon hits. About 8% of excerpts retain a detectable place or school name; the stance–partisanship gradient is present in excerpts without one (random +1.16 per 10 pp, $p < .001$) and weaker where one remains (+0.59, $p = .05$), so leakage does not generate the result.

Attribute definitions (version 3). *Hostility*: personal attacks or insults, shouting or interrupting, accusations of lying, corruption or bad faith, threats, calls for officials to resign or speakers to be removed; polite disagreement is low. *Culture-war share*: share of the excerpt on race in the curriculum, library or classroom content, gender identity, parental rights, masks or vaccines; routine business counts as none. *Parental-rights framing*: how strongly issues are framed as parents' authority (right to know, consent, opt out) rather than the district's. *COVID contention*: disagreement over masks, closures or remote learning, vaccines, quarantines, testing; neutral updates are zero. *Progressive / conservative positions*: how strongly speakers argue that side of a contested dispute, with examples (defending books against removal, DEI or LGBTQ-inclusive policy, mask requirements; demanding removal of books, opposing DEI, asserting parental authority over curriculum, opposing CRT, opposing mask requirements); positions argued, not topics discussed; routine descriptions of district programs score zero. Version 2 lacked the last clause: in the pilot it scored a staff equity-plan presentation 57 on progressive positions, and random excerpts averaged 4.4 progressive against 0.3 conservative. Version 3 was re-piloted on the same 200 split-A excerpts and frozen. Every wording tried is logged in `source/analysis/gabriel/attribute_log.md`.

Validation. Scoring used `gpt-6-luna` with reasoning disabled, JSON output and 10 parallel calls; total API cost was $16.93. Pilot run-to-run correlations on 200 excerpts: 0.97, 0.98, 0.97, 0.99, 0.96, 0.99 for the six attributes in the order above, with mean absolute differences of 0.6–3.3 points. Re-scoring 1,000 excerpts (500 random, 500 flagged) with `gpt-6.1-terra`

gives correlations of 0.95, 0.93, 0.95, 0.98, 0.85, 0.97; Terra scores progressive positions 2.4 points lower on average, applying the routine-description rule more strictly, with the largest disagreement in random excerpts ($r = 0.70$) where most scores are near zero. We therefore lead with conservative positions ($r = 0.97$) and report stance (conservative minus progressive) only as a secondary measure. Post-period regressions re-estimated on the 1,000 shared excerpts give similar coefficients with either model's scores.

Public-comment share. A one-attribute pass over the 22,174 original and extension excerpts asked what share of each is public comment by members of the public, as opposed to board or staff discussion, using cues such as "state your name and address," "three minutes" and "next speaker" (median 0, mean 12.3). Excerpts with share ≥ 50 are 5–10% of random excerpts by year (9.7% in 2021; 199 excerpts from 149 districts). The extra 2021 and 2023 random windows were not scored for public comment and are excluded from the by-speaker split (Fig. 12).

Quotes. For Indian River (FL, Republican share 0.61) and Radnor Township (PA, 0.36), chosen from the three largest pre-to-post risers in each group, the 25 highest-scoring excerpts on the relevant position attribute and the 10 highest on hostility were passed in raw form to `gpt-6.1-terra`, which returned the single sentence most strongly arguing the position. A quote is kept only if it is an exact substring of the raw window (34 of 35 and 26 of 35 kept; failures are excerpts with no sentence returned). Private individuals' names are replaced afterwards; caption errors are preserved. Full lists are in `output/tables/quotes_1200930.csv` and `quotes_4219920.csv` in the project repository.